LITERATURE SURVEY

This work is dedicated to the design and perpetration of a washing machine using Verilog on vivado. A washing machine is a ménage appliance used to wash clothes. The term is substantially used for machines that use water as opposed to dry cleaning or ultrasonic cleansers. The stoner adds soap, which is vended in liquid or greasepaint form, to the washing water. Bendix Home Appliances, a attachment of AVCO, introduced the first home automatic washing machine in 1937. The foremost washing machines simply did the washing when loaded with clothes and cleaner, filled with hot water and run. Over time, the machines came more and more automated, first with veritably complex electromechanical controls, also completely electronic controls into which druggies would fit apparel. machine, elect the applicable program with a switch, and this program is written in HDL(tackle description language) law, as we use verilog then, and return to remove the clean and slightly damp laundry at the end of the cycle. In this design, the design of an automatic washing machine is carried out. The design is enforced and synthesized at the register transfer position(RTL) using the tackle Description Language Verilog. perpetration and simulation is done using XILINIX VIVADO(22.2)

INTRODUCTION

- Various day to day life situations can be represented by FSM (finite state machine) like automated washing machine.
- FSM is an theoretical machine that can be in exactly one of the finite number of States at any given instant of time, the FSM can shift from one state to another on response to some inputs, this change is called Transition.
- An FSM is defined by a list of its states, it’s initial state and the inputs that triggers each transition
- Here we allocate different stages of washing machine such as add water, add detergent, wash, drain etc.
- Now we will see how this FSM operates by writing testbench.

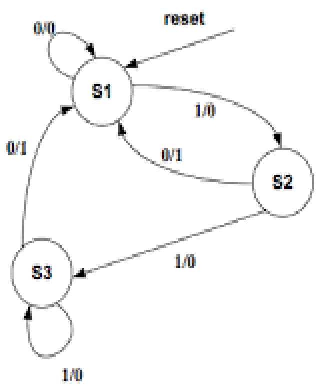

Fig 1 : FSM

Types of FSM

- **Mealy FSM**

  The state machine whose output values are depends upon both by its current state and the current state.

- **Moore FSM**

  The State machine whose output values depends upon solely by its current states.

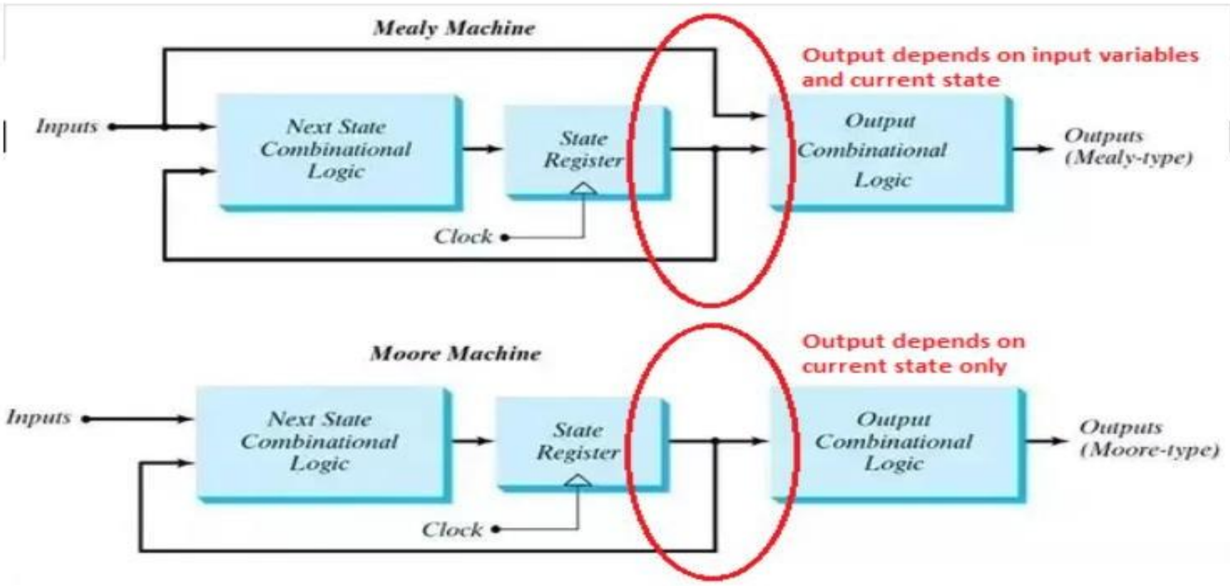

Fig 2 : Mealy vs Moore Machine

Working of Washing Machine

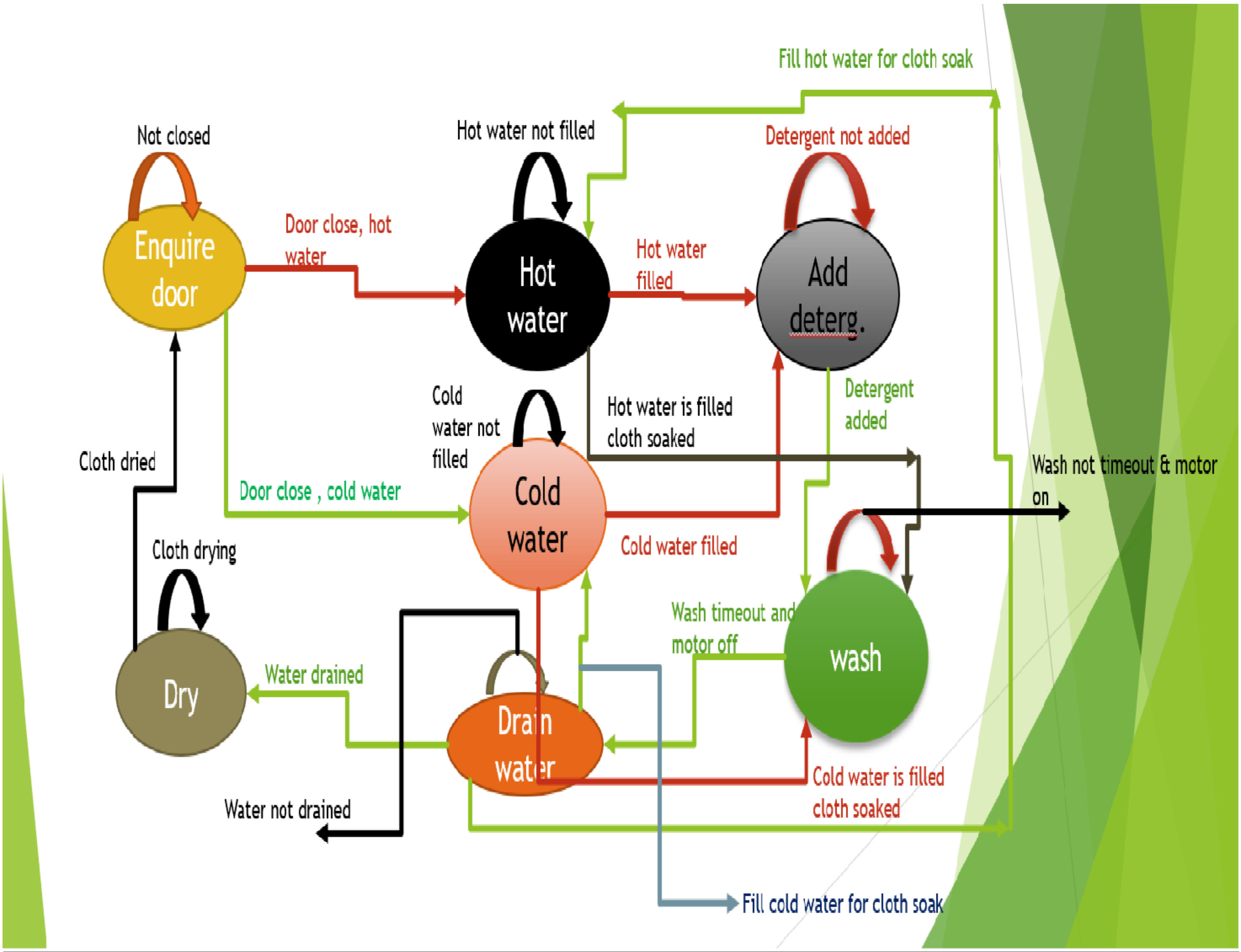

Fig 3 : State Diagram of Mealy FSM of Automatic Washing machine

- As shown in above state diagram the washing machine is a mealy FSM with seven stages enquire door , hot water , cold water ,add detergent , wash ,drain water ,dry .
- Now if door not closed then it remain on the current state which is enquire door but when door get closed then if input is hot water moves to state or otherwise move to cold water state.
- When it is in hot water state then if not filled with hot water remain in current state else move to add detergent state.
- When it is in cold water state then if not filled with cold water remain in current state else move to add detergent state.
- When it is in add detergent state until detergent is not added remains on this state only but when the detergent is added sufficiently move to state wash.
- In wash state the machine remains until wash is not timed out and motor is on but move to drain water state when wash is timed out and motor turns off.
- In drain water state it remain on that state unless water is not drained but what happens if in this draining of dirty water some useful hot or cold water got drained then it again to hot water state if input is fill hot water for soaked cloth or goes to cold water state if input is fill cold water for soaked cloth and goes to dry state if water is drained after this all process.
- When it goes to hot water state again then again hot water is filled and goes directly to wash state with input as hot water is filled cloth soaked skipping add detergent state.
- When it goes to cold water state again then again cold water is filled and goes directly to wash state with input as cold water is filled cloth-soaked skipping add detergent state.
- After all this when state arrives at dry state it remains there until input is cloth drying but moves to starting state and complete its total flow when input of cloth dried comes.

Verilog Code

```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
// Create Date: 18.01.2025 :23:57
// Design Name:
// Module Name: min project
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
// Dependencies:
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//////////////////////////////////////////////////////////////////////////////////


module min_project(
clk , reset , hotwater , coldwater , dr_cl , start , filled , detergent_added , wash_time_out ,
drained , dry_time_out , soap_wash_check , door_lock,motor_on , hot_fill_value_on ,
cold_fill_value_on ,
drain_value_on , completed , soap_wash , cold_water_wash , hot_water_wash
);
input  clk ,  reset ,  hotwater ,  coldwater  ,  dr_cl  ,  start  ,  filled  ,  detergent_added  ,
wash_time_out,drained  ,  dry_time_out  ,  soap_wash_check;

output reg door_lock  ,  motor_on  ,  hot_fill_value_on  , cold_fill_value_on  ,
drain_value_on  ,  completed ,  soap_wash  ,  cold_water_wash  ,  hot_water_wash;

//defining the states

parameter enquire_door=3'b000;
parameter hot_water=3'b001;
```

```
parameter cold_water=3'b010;
parameter add_deter=3'b011;
parameter wash=3'b100;
parameter drain_water=3'b101;
parameter dry=3'b110;
reg [2:0] present_state,coming_state;
always@(posedge clk )
begin
if(reset)
begin
present_state=3'b000;
end
else
begin
present_state<=coming_state;
end
end
always@(*)
begin
case(present_state)
3'b000:if(start==1'b1&&dr_cl==1'b1&&hotwater==1'b1)
begin
coming_state=hot_water;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b0;
cold_water_wash=1'b0;
hot_water_wash=1'b0;
completed=1'b0;
end
else if(start==1'b1&&dr_cl==1'b1&&coldwater==1'b1)
begin
```

```
coming_state=cold_water;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b0;
cold_water_wash=1'b0;
hot_water_wash=1'b0;
completed=1'b0;
end
else
begin
coming_state=present_state;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b0;
soap_wash=1'b0;
cold_water_wash=1'b0;
hot_water_wash=1'b0;
completed=1'b0;
end
3'b001:if(filled==1'b1)
begin
if(soap_wash_check==1'b0)
begin
coming_state=add_deter;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;
```

```
cold_water_wash=1'b0;
hot_water_wash=1'b0;
completed=1'b0;
end
else
begin
coming_state=wash;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b0;
cold_water_wash=1'b0;
hot_water_wash=1'b1;
completed=1'b0;
end
end
else
begin
coming_state=present_state;
motor_on=1'b0;
hot_fill_value_on=1'b1;
cold_fill_value_on=1'b0;

door_lock=1'b1;

completed=1'b0;
end

3'b010:if(filled==1'b1)
begin
if(soap_wash_check==1'b0)
begin
coming_state=add_deter;
```

```
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;
cold_water_wash=1'b0;
hot_water_wash=1'b0;
completed=1'b0;
end
else
begin
coming_state=wash;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b0;
cold_water_wash=1'b1;
hot_water_wash=1'b0;
completed=1'b0;
end
end
else

begin
coming_state=present_state;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b1;

door_lock=1'b1;

completed=1'b0;
```

```
end
3'b011:if(detergent_added==1'b1)
begin

coming_state=wash;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;

completed=1'b0;
end
else

begin
coming_state=present_state;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;

completed=1'b0;
end
3'b100:if(wash_time_out==1'b1)

begin
coming_state=drain_water;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
```

```
door_lock=1'b1;

completed=1'b0;
end
else
begin
coming_state=present_state;
motor_on=1'b1;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
completed=1'b0;
end
3'b101:if(drained==1'b1)
begin
if(hot_water_wash==1'b0)

begin
coming_state=hot_water;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;

completed=1'b0;
end
else if(cold_water_wash==1'b0)

begin
coming_state=cold_water;
motor_on=1'b0;
hot_fill_value_on=1'b0;
```

```
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;

completed=1'b0;
end
else
begin
coming_state=wash;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b0;
cold_water_wash=1'b1;
hot_water_wash=1'b0;
completed=1'b0;
end

begin
coming_state=dry;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;
cold_water_wash=1'b1;
hot_water_wash=1'b1;
completed=1'b0;
end
end
else
```

```
begin
coming_state=present_state;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b1;
door_lock=1'b1;
soap_wash=1'b1;

completed=1'b0;
end
3'b110:if(dry_time_out==1'b1)

begin
coming_state=enquire_door;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b0;
door_lock=1'b1;
soap_wash=1'b1;
cold_water_wash=1'b1;
hot_water_wash=1'b1;
completed=1'b1;
end
else

begin
coming_state=present_state;
motor_on=1'b0;
hot_fill_value_on=1'b0;
cold_fill_value_on=1'b0;
drain_value_on=1'b1;
door_lock=1'b1;
```

```
soap_wash=1'b1;
cold_water_wash=1'b1;
hot_water_wash=1'b1;
completed=1'b0;
end
default:coming_state=enquire_door;
endcase
end
endmodule
```

Test Bench

```
`timescale 10ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 18/01/2025
// Design Name:
// Module Name: akt
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////



//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10.01.202512:24:52
// Design Name:
// Module Name: tb
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
```

```
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////


module testbench;
    reg clk,  reset, hotwater, coldwater,dr_cl ,start,
filled,detergent_added,wash_time_out,drained,dry_time_out,soap_wash_check;
    wire door_lock,motor_on,hot_fill_value_on,cold_fill_value_on,
drain_value_on,completed,soap_wash,cold_water_wash,hot_water_wash;


     min_project
DUT1(clk,reset,hotwater,coldwater,dr_cl,start,filled,detergent_added,wash_time_out,drai
ned,dry_time_out,soap_wash_check,door_lock,

motor_on,hot_fill_value_on,cold_fill_value_on,drain_value_on,completed,soap_wash,co
ld_water_wash,hot_water_wash);



  initial
    begin
    clk=1'b0;
     reset=1'b1;
     dr_cl=1'b0;
     hotwater=1'b0;
     coldwater=1'b0;
     start=1'b0;
     filled=1'b0;
     detergent_added=1'b0;
     wash_time_out=1'b0;
```

```
drained=1'b0;
dry_time_out=1'b0;
soap_wash_check=1'b0;

#5 reset=1'b0;
#5  start=1'b1;dr_cl=1'b1;

#10 hotwater=1'b1;
#10 coldwater=1'b1;
#10   filled=1'b1;
#10   detergent_added=1'b1;
#10   wash_time_out=1'b1;
#10   soap_wash_check=1'b1;
#10  drained=1'b1;
#10
dry_time_out=1'b1;
end
always
begin
#5 clk=~clk;
end
endmodule
```

Simulation Waveform

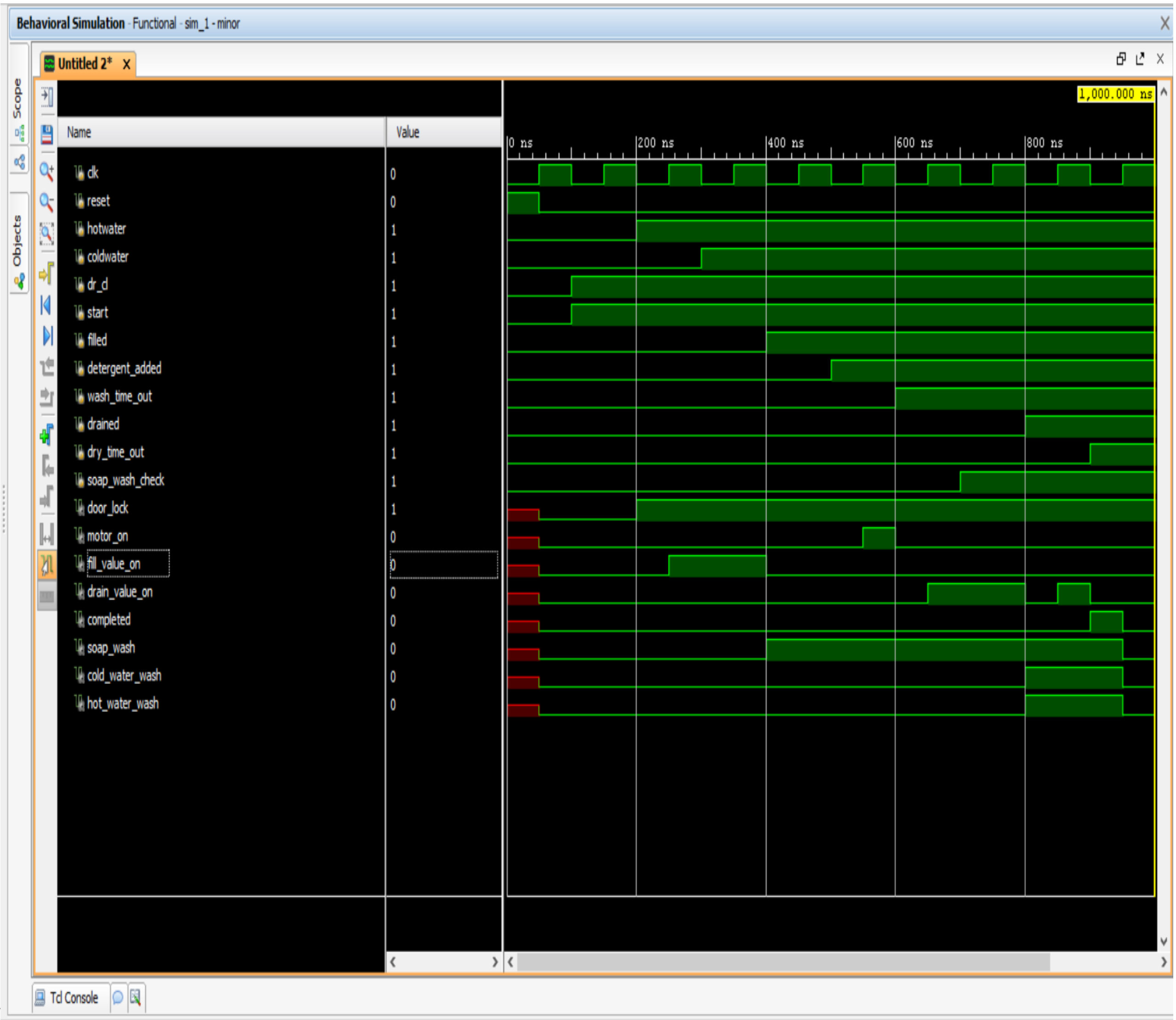

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Fig 4 : Washing Machine I/O simulation result

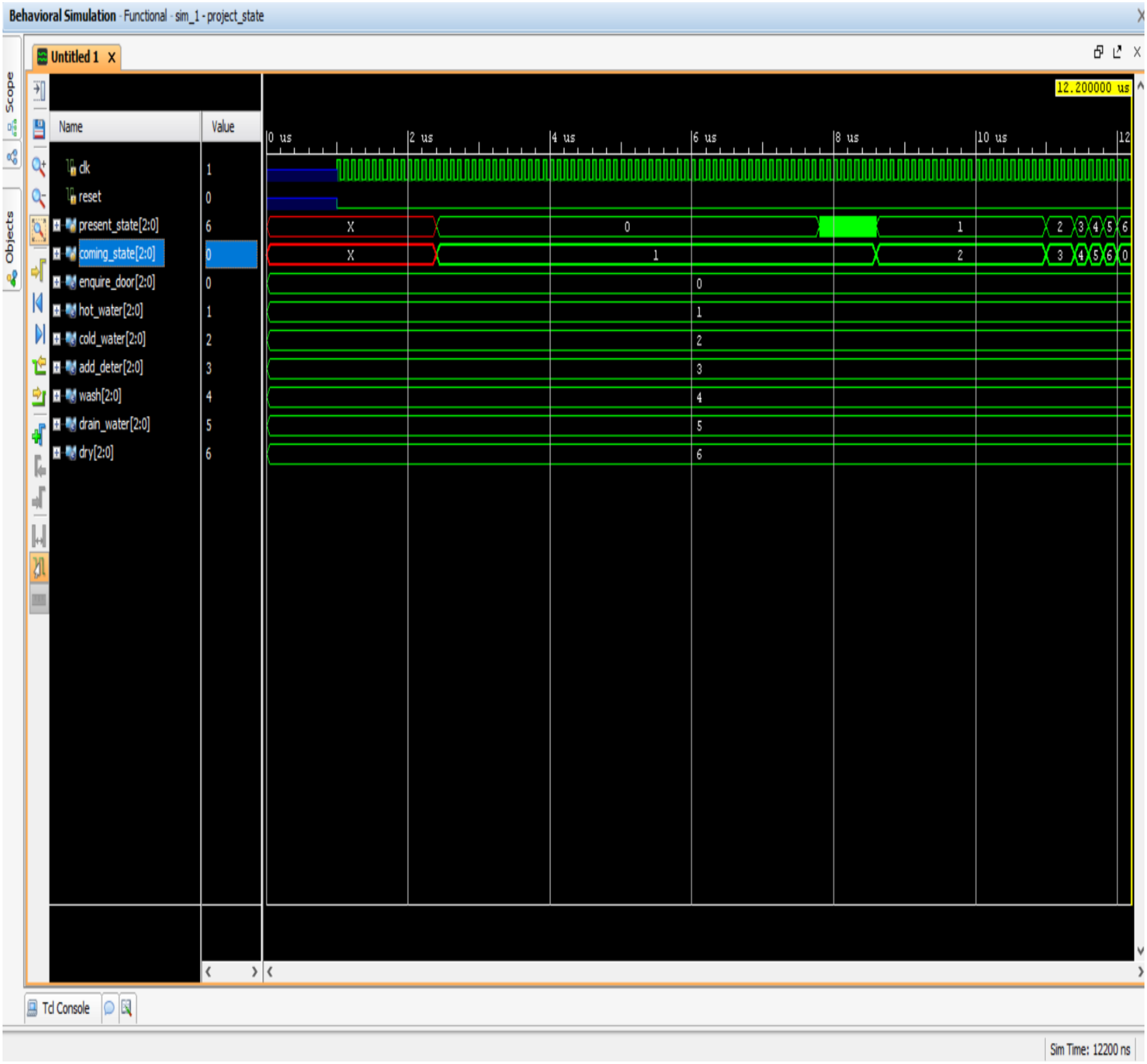

Fig 5 : Washing machine present and next state simulation waveform

Observation of Timing Diagram

- Firstly, the reset is high so all the outputs are in high impedance state after then reset is set low then machine start operating.
- In the initial part of simulation itself I have turn on the start and the door closed variable as the input.
- Then set the inputs high for add detergent and water filled.
- Based on different inputs the outputs also changed.
- The door closed named as door_lock is input is set high as soon as the machine is started and the door is closed and the motor is turned on.
- The water is filled , the water is drained out.
- As soon as all the edges are done , the complete variable which is output is finally high which indicate that the operation have been completed.

RTL Schematic

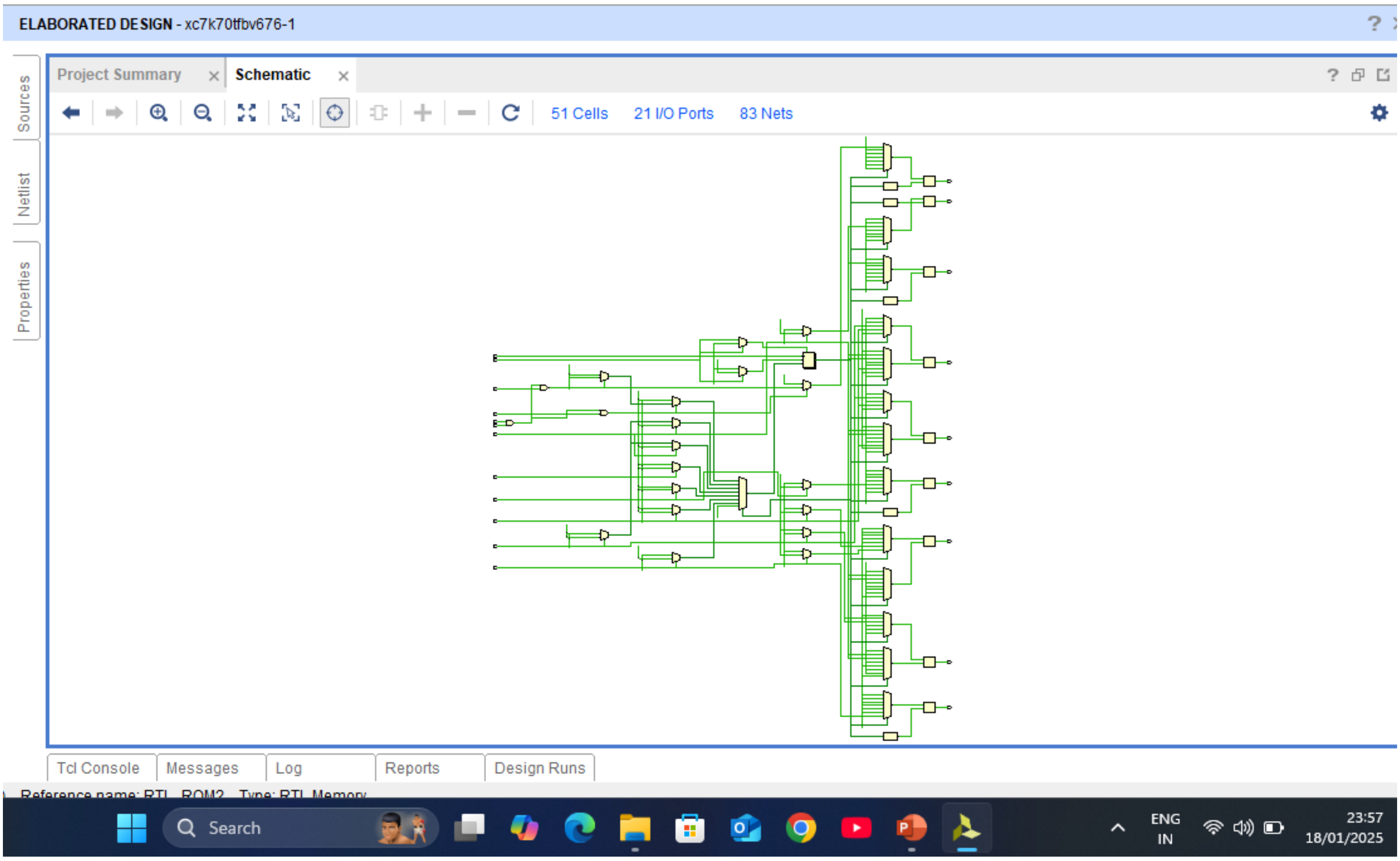

Fig 6 : RTL Schematic of washing Machine

Conclusion & Future Scope

In this project, we have successfully implemented washing machine FSM using Verilog on vivado tool and its RTL design and timing simulation is shown.

This project is based on FSM, various another project can be developed on this FSM.
More functionality may be included in this project like different mode to wash the clothes based on clothing materials, temperature etc.

References

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2. https://rashmistudents.yolasite.com/resources/Verilog%20HDL%20-%20Samir%20Palnitkar.pdf
3. https://nptel.ac.in/courses/106105165
4. http://ebook.pldworld.com/_eBook/FPGA%EF%BC%8FHDL/-Eng-/Verilog%20digital%20systems%20design%20(Navabi).pdf